

Please refer to pages **8–21** of the recommended text book.

UNIT 1.3

WHAT IS FUNCTION?

Learning outcomes: On completion of this unit you should

- (1) be able to identify a function.
- (2) be familiar with the four possible ways of representing a function.
- (3) be able to find the domain of a function.
- (4) be able to find the range of some functions by using the graph of the function.
- (5) know how to test whether a curve in the xy –plane is the graph of a function by using the Vertical line test.
- (6) be familiar with piecewise-defined functions.
- (7) be able to sketch the function $y = |f(x)|$ if the graph of the function $y = f(x)$ is given.
- (8) be able to write down and use the definition of an even function.
- (9) be able to write down and use the definition of an odd function.

- (10) be able to write down and use the definition of an increasing function.
- (11) be able to write down and use the definition of a decreasing function.

Definition of a function. (please refer to page 8 of the recommended text book.)

Representing functions. (please see page 10–12 of the text book)

Verbally: by description in words

Numerically: by a table of values

Visually: by a graph

Algebraically: by an explicit formula

Finding Domain and range of a function.

Consider the following functions

$$(1) f(x) = \sqrt{1-x}$$

$$(2) g(t) = \frac{1}{t^2}$$

$$(3) A(r) = \pi r^2$$

$$(4) \ h(u) = 5^u$$

Piecewise defined functions (see page 14).

Consider the following functions and discuss their domain and range

$$(1) \ f(x) = \begin{cases} 4, & \text{if } x < -2 \\ -x^2 + 1, & \text{if } -2 \leq x < 1 \\ x + 4, & \text{if } x > 1 \end{cases}$$

$$(2) \ h(x) = x + |x + 1|$$

Sketching graph of $y = |f(x)|$ from the graph of $y = f(x)$.

Consider the following examples

(1) $y = |x + 1|$

(2) $y = |x^2 + 1|$

(3) $y = |-2^x|$

Tips to Identify rules which define functions (see page 13).

The Vertical line test:

Determine whether or not $y = f(x)$ defines a function

- $x^2 + y^2 = 4$

- $x = y^2$

Even and Odd functions or Symmetric functions (See page 16).

Even functions: Symmetric with respect to the y -axis

Odd functions: Symmetric with respect to the origin

Increasing and decreasing functions.

Increasing functions:

Decreasing functions: